

# Kayla Ippongi

kaylaipp.github.io - kaylasippongi@gmail.com - linkedin.com/in/kaylaipp

## Education

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**Johns Hopkins University**, M.S, Artificial Intelligence

2021 – 2024

Deep Neural Networks, Intelligent Algorithms, Creating AI Enabled Systems, Deep Learning for Computer Vision, Cognitive & Behavioral Foundations for Artificial Intelligence, Machine Learning, Artificial Intelligence, Foundations of Algorithms, Multivariable Calculus

**Boston University**, B.A, Computer Science

2016 – 2019

Computer Systems & Architecture, Data Structures & Algorithms, Artificial Intelligence, Data Science in R, Linear Algebra, Data Mining, Data Mechanics, Data Science in Python, Concepts of Programming Languages, Intro to Databases, Intro to Computer Graphics

## Technical Skills

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### Languages & Web frameworks:

Python, TypeScript, GraphQL, React, Java, NodeJS, Jest, Flask, Django

### AI / ML Frameworks & Libraries:

Tensorflow, Keras, Scikit-Learn, Pandas, NumPy

### Infrastructure/Systems:

AWS (Lambda, StepFunctions, DynamoDB), Serverless, Terraform, DataDog, Grafana, MongoDB, Elasticsearch, Docker, Kubernetes, k6 Load Testing

## Experience

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**Rivian** | Senior Software Engineer

Palo Alto, CA | Sept 2022 – Present

- Optimized legacy vehicle service pricing API latency from 15s to 1.5s (10x) by analyzing performance bottlenecks using **DataDog** APM and distributed tracing, saving an estimated \$6M/year in wait time savings. Supported with integrated **CI/CD k6 load tests** to enforce performance budgets and validated improvements with production observability metrics post-rollout
- Led integration of 3rd party SOAP platform into our **Typescript, GraphQL** backend and **React** frontend, to automate Fleet authorization workflows, which replaced manual processes, reducing approximately 131 labor hours/day, and \$2M annually in operational savings
- Migrated customer e-signature services and APIs from OneSpan to Box using **Typescript, GraphQL, and AWS Lambda**, ensuring feature parity, seamless customer experience, and long-term backend scalability, resulting in ~\$800K/year in operational savings
- Led technical design discussions and authored architecture and trade-off documentation for new services and larger system changes with cross-functional stakeholders and engineers

**Rivian** | Software Engineer

Palo Alto, CA | April 2021 – Sept 2022

- Designed and implemented backend services for Rivian's Recall and Quality Containment dashboard using **Typescript, GraphQL, AWS Lambda, DynamoDB, StepFunctions** for batch orchestration, reducing manual tracking efforts and accelerating issue resolution across thousands of serviced vehicles
- Improved recall data retrieval speed by 6x by leveraging **ElasticSearch**, implementing indexing and querying strategies for high-volume data processing
- Triaged production incidents using metrics, distributed traces, and observability dashboards, performed root-cause analysis, and implemented fixes to improve service reliability and reduce risk of regression

**Wayfair** | Software Engineer

Boston, MA | July 2019 – Feb 2020

- Developed and maintained features on the Storefront Product Options team using **PHP, GraphQL, and React**, improving consumer engagement and product visibility across millions of products.
- Collaborated cross-functionally to remove service dependencies and migrate teams to our new API, enhancing system scalability and reducing operational complexity
- Implemented multivariate and A/B feature tests for Wayfair and its three partner sites, leveraging our lead image algorithm, to determine best images across 20 different product classes in preparation for the company's largest sale day
- Refactored and optimized **stored SQL procedures** to improve backend query performance, reducing latency for production option data retrieval across thousands of live products.

**IBM Cloud** | DevOps Developer Intern

Dallas, TX | May 2018 – July 2018

- Developed and deployed a REST API for the Cloud Operations security team to help optimize Kubernetes port lookup process for internal services. Built utilizing **Python** Kubernetes client, **Flask, Kubernetes, Docker** and **Minikube**

## Projects

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**ASL Isolated Gesture Recognition (Github)**: Built a custom CNN using TensorFlow/Keras to classify sign language gestures from video frame data, implementing data preprocessing, frame sampling, and multi-class model training to establish a strong baseline accuracy

**YOLOv4 Street Parking Detection (link)**: Curated and annotated 4.5K frames across Argo, Waymo, and nuScenes; trained YOLOv4 vs. Faster R-CNN for parking detection, achieving 83% precision and demonstrating advantages of cross-dataset training (mAP ≈ 64%)